l CHAPTER 1: THE PROJECT AND ITS BACKGROUND

This chapter contains the general overview of the study. It presents the problem addressed by the research, identifies the client and their requirements, and describes the proposed project or solution. The chapter also states the project objectives, discusses the scope and delimitation of the study, and the design constraints considered during development. It outlines the applicable engineering standards and explains the engineering design process conducted by the researchers

1.1 The Problem

The Bachelor of Science in Computer Engineering program at Pangasinan State University – Urdaneta City Campus requires students to complete On-the-Job Training (OJT) as a critical component of their curriculum. This practicum period helps students connect what they learn in the classroom with real work in the industry. It lets them practice their skills, gain experience, and use their technical knowledge in real situations.. However, significant challenges persist in managing and monitoring student internships effectively.

Traditional OJT management at PSU–Urdaneta City Campus relies heavily on manual, paper-based processes. Coordinators and instructors utilize physical logbooks, printed timesheets, and confusing communication channels such as email and messaging applications to track student progress. These conventional methods introduce critical inefficiencies that hinder the effectiveness of the internship program.

Relying on physical documentation creates risks of data redundancy and loss. Essential records including Daily Time Records (DTR), evaluation forms, and internship agreements can be easily misplaced, damaged, or destroyed, resulting in permanent loss of critical academic documentation.This weakness endangers the accuracy of student records and adds extra administrative work when lost data must be restored.

Likewise, delayed feedback mechanisms also become a real problem. The manual submission and review of reports and evaluations introduce significant time lags between student performance and instructor response. This delay prevents instructors from providing timely support to students who are struggling with their internship tasks, which may affect their learning outcomes and professional development.

Moreover, Attendance verification shows another difficulty in the current system. Without real-time tracking mechanisms, coordinators and instructors face difficulties in confirming the actual presence of students at their deployment sites.This problem can cause differences between the hours students report and the time they actually spend on internship tasks, making the records less accurate.

The administrative burden on faculty members represents a major inefficiency in the current system. Coordinators and instructors allocate unequal time to manually gathering data for compliance reports, time that could be better invested in mentorship and student guidance. Excessive administrative tasks reduce the level of supervision and support available to students during their internship.

Document security and accessibility are also concerns in the paper-based system. Storing confidential student documents in physical form raises concerns about data privacy, authorized access, and long-term preservation. Without centralized and secure storage, it is difficult to follow proper document retention policies and meet data protection requirements.

These challenges demonstrate the need for a modernized, digital solution to OJT management at PSU–Urdaneta City Campus.Inefficiencies in manual processes affect all stakeholders: students experience delayed feedback, instructors spend too much time on administrative tasks, and coordinators face difficulties keeping accurate and complete internship records.

1.2 The Client

The primary client for this project is the Computer Engineering Department of Pangasinan State University – Urdaneta City Campus. The department manages the academic and professional development of Computer Engineering students, including the supervision of their On-the-Job Training programs. The Computer Engineering Department is a key stakeholder in this project, with specific needs for managing internships. The department supervises multiple student cohorts across various industry partners, requiring efficient tracking systems, detailed reporting, and clear communication between all parties involved.

Within the department, different user roles participate in managing internships, each with specific responsibilities and system needs. The OJT Coordinator serves as the primary administrator, responsible for overall program oversight, student assignment to companies, document validation, and compliance reporting. The coordinator requires comprehensive access to all system functions to fulfill these duties effectively.
The students will use the modernized system to submit documents, record attendance, track progress, and receive feedback from supervisors. Their experience with the system directly affects their engagement in the internship program and adherence to submission requirements.

Industry partners, specifically assigned supervisors at internship host companies, form an external client group with limited interactions with the system. These supervisors are able to verify student attendance, submit performance evaluations, and provide feedback on intern contributions. Their participation in the system ensures accurate assessment of student performance in actual workplace environments.

The client's ultimate objective centers on enhancing the quality, efficiency, and transparency of the OJT program for Computer Engineering students. By implementing a centralized digital platform, the department aims to reduce administrative overhead, improve feedback timeliness, ensure accurate record-keeping, and ultimately provide a superior internship experience that effectively prepares students for professional engineering careers.

1.3 The Project/Solution

The INTRAK system represents a comprehensive web-based solution designed specifically to address the internship management challenges faced by the Computer Engineering Department at PSU–Urdaneta City Campus.The system offers a centralized digital platform that streamlines the entire internship process, from student deployment to final evaluation and certification. It also incorporates a centralized storage to secure data management.The system is deployed on cloud infrastructure, ensuring compliance with university data policies and providing secure, role-based access to authorized users including students, instructors, coordinators, industry partners and administrators.

The system completes disjointed paper-based processes with a unified digital workflow for document submission, review, approval, and storage. Students upload required internship documents directly through the web interface, where they are automatically routed to appropriate reviewers based on document type and student assignment. This automated workflow eliminates manual document routing and reduces processing delays inherent in physical document handling.

The researchers implement role-based access control in the system to ensure appropriate data privacy and functional access for each user category. Coordinators have extensive program-level administrative capabilities, including managing students, assigning students to instructors, managing company partnerships and MOAs, reviewing documents for all students, verifying attendance, and generating comprehensive reports. Instructors can access functions for their assigned student cohorts, such as reviewing documents, verifying attendance, submitting evaluations, managing document templates, and reviewing company applications. Students use self-service functions to upload documents, log attendance, apply to companies, track their progress, and view evaluations. Industry supervisors access limited functions focused on attendance verification, evaluation submission (Supervisor Feedback Form), and document review for their assigned interns.

Attendance tracking has multiple verification features to support different internship settings. Students record their daily attendance through the web interface, creating timestamped records of their internship activities. The system keeps detailed logs of time-in, time-out, and total hours, allowing instructors to track progress toward required hours.

Document management goes beyond basic file storage to include tracking document status, managing review workflows, and providing feedback. Each submitted document progresses through various approval states such as pending, approved, rejected, or resubmission requested. Both instructors and coordinators can provide structured feedback on document submissions, which helps improve the overall quality of internship documentation. Additionally, the system includes a direct messaging feature that enables coordinators, students, and instructors to communicate directly within the system. Students can message their assigned instructors and coordinators, instructors can message their assigned students, and coordinators can message any student, facilitating real-time communication and collaboration throughout the internship process.

The evaluation system facilitates structured performance assessments from industry supervisors. Standard evaluation forms collect both numerical ratings and written feedback on various performance aspects, producing complete performance profiles for each student. As the INTRAK system gathers data from students (such as attendance logs and document submissions) and receives evaluations from industry supervisors, the evaluation data supports interns through counseling and other program improvement initiatives.

The reporting features allow coordinators to generate compliance documents, monitor program metrics, and identify students needing support. The system produces attendance summaries, document status reports, evaluation summaries, and completion records. These reports assist stakeholders such as academic administration, accreditation bodies, and industry partners.

The chosen web-based deployment model is used for easy access of various devices in different locations while maintaining security with different controls. The responsive interface adapts to different screen sizes, though mobile application development falls outside the current project scope. Users access the system through standard web browsers, avoiding client software installation and streamlining system management.

INTRAK transforms OJT management at PSU–Urdaneta City Campus from a manual, paper-intensive process into a streamlined, digital workflow. By centralizing information, automating routine tasks, and providing real-time visibility into student progress, the system enables the department to deliver a higher quality internship experience while reducing administrative burden on faculty members.

1.4 The Project Objectives

The primary objective of this study is to develop and implement INTRAK, a centralized, web-based Internship Tracking and Evaluation System with an integrated Raspberry Pi–based Network-Attached Storage (NAS), designed to improve the management, monitoring, documentation, and evaluation of On-the-Job Training (OJT) activities of

Computer Engineering students at Pangasinan State University – Urdaneta City Campus.

This project aims to:

1. Centralize internship tracking and documentation by replacing manual and fragmented record-keeping processes with a unified digital platform that ensures accuracy, consistency, and accessibility of internship data.

2. Enable efficient digital submission and management of internship requirements, including Daily Time Records (DTR), Memoranda of Agreement (MOA), endorsement letters, performance evaluations, and other required documents through a user-friendly web interface.

3. Provide coordinators and instructors with validation, monitoring, and evaluation tools that support document review, approval, feedback provision, and systematic assessment of student performance while reducing administrative workload.

4. Implement a secure, login-based attendance and activity logging mechanism that records time-stamped internship hours, ensuring transparency, accountability, and reliable verification of student participation.

5. Generate automated reports and analytics to support student assessment, coordinator supervision, and administrative decision-making, thereby minimizing time spent on manual report preparation.

6. Integrate a Raspberry Pi–based NAS solution for secure, centralized, and institution-controlled storage of all internship-related documents, ensuring data security, controlled access, and long-term retention without dependence on external cloud services.`

1.5 Scope and Delimitation

This section states the scope and delimitations of the study by defining the system coverage, users, and technologies involved in the development of the INTRAK system and its implementation constraints.

1.5.1 Scope

The project focuses on the development of INTRAK: A Centralized Internship Tracking and Evaluation System with NAS-Based Document Storage, specifically designed for Computer Engineering students of Pangasinan State University – Urdaneta City Campus. The system aims to streamline the monitoring, evaluation, and documentation of student internship activities. Specifically, the system will:

1. Serve as a centralized platform for tracking internship activities and document submissions of Computer Engineering students.

2. Allow students to submit digital copies of internship requirements, including Daily Time Records (DTR), performance evaluations, and other related documents.

3. Enable coordinators to validate student submissions, provide feedback, and generate reports.

4. Monitor attendance through secure date and time logs of student logins.

5. Provide coordinators with evaluation and performance assessment tools to monitor intern progress.

6. Generate reports for students, coordinators, and the university to support documentation and assessment requirements.

7. Utilize a Network-Attached Storage (NAS) for secure, centralized storage and controlled access to internship-related documents.

8. Cover exclusively the Computer Engineering Department of PSU–Urdaneta City Campus.

9. Use login-based validation as the primary attendance tracking method; real-time geolocation tracking will not be implemented, focusing instead on timestamp-based verification.

1.5.2 Delimitation

This section presents the delimitations of the study as identified by the researchers. The study is limited to the following parameters:

1. The system will not initially support ISO-compliant e-signatures; physical signatures may still be required for official documents.

2. The study is limited to Computer Engineering students and coordinators only; other departments are not included

3. Hard copy submission of certain requirements may still be required as back up depending on the University policy.

4. The system will be web-based only; no mobile application will be developed within this study.

1.6 Design Constraints

This section identifies the design constraints considered in developing the INTRAK system to ensure it operates within technical, resource, and operational limitations.

1.6.2 Operational Constraints

The INTRAK system must follow university policies on document retention, academic record-keeping, and data privacy, which place limits on what the system can do. Some official internship documents still require physical signatures and must be kept as hard copies. Because of this, the system cannot be fully digital and must support a combination of digital and physical processes, known as a hybrid workflow. The academic calendar and available resources also limit the system's development timeline. Only essential features can be implemented in the initial version. More advanced features, such as ISO-compliant digital signatures or automatic integration with external company HR systems, are not included in this first release due to time constraints.

User training and change management are also identified as operational considerations. Since students, instructors, and coordinators may be used to paper-based processes, the system must have simple and intuitive interfaces. This ensures that users can easily navigate the system, submit documents correctly, and monitor internship progress without needing extensive training. Designing the system in this way also helps reduce resistance to change and encourages adoption across all user groups.

1.6.3 Economic Constraints

The project's budget limits the choice of hardware, software tools, and ongoing operational costs. The researchers can use the Raspberry Pi as a Network-Attached Storage (NAS) device or commercial NAS device instead of relying on cloud storage, reducing recurring subscription costs. This one-time investment aligns with budget constraints while still providing sufficient storage capacity for all internship documents.

The system uses open-source technologies including PostgreSQL, React, and Node.js also helps address economic limitations. These platforms are free to use, well-supported, and reliable, which allows the development of a professional-quality system without spending extra funds on software licenses. By choosing cost-effective solutions, the project can remain within budget while maintaining system quality and reliability.

1.6.4 Environmental Constraints

The system's hardware, including the Raspberry Pi NAS, must be energy-efficient to minimize electricity consumption. Selecting hardware that uses less power helps the university save energy and supports institutional sustainability goals.

Also, moving from paper-based to digital workflows reduces environmental impact. Less paper is needed, and there is a smaller need for physical storage and transportation of documents. This shift contributes positively to the environment and aligns with sustainable practices encouraged by the university

1.6.5 Security and Privacy Constraints

The system must follow the Data Privacy Act of 2012 and university policies, which impose rules on how student data is stored, accessed, and monitored. Only authorized users, such as coordinators and instructors, should have access to sensitive information, while students should only see data relevant to them. This ensures that personal information is protected from unauthorized access or misuse.

Restricting system access to Computer Engineering Department users simplifies security management. It creates a clear boundary of responsibility, ensures that all data is managed under a single department, and reduces the risk of data breaches. However, this restriction also limits the ability to integrate the system across other departments or units in the university.

1.7 Engineering Standards

The development, implementation, and operation of INTRAK adhere to established engineering standards that ensure quality, security, and maintainability throughout the system lifecycle. This section contains the engineering standards followed in this study.

1.7.1 Software Quality Standards

ISO/IEC 25002:2024 -Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE)

This document uses a quality model to describe how the INTRAK system is evaluated and improved. The quality model serves as a guide for defining requirements, checking whether these requirements are met, and understanding how the system performs in different areas. The sections below present the main quality characteristics considered in the development and assessment of INTRAK.

Functional Suitability

This characteristic explains how well the system supports the needs of its users. For INTRAK, it involves making sure that all required functions for OJT management are present and working as intended, based on the needs identified by the Computer Engineering Department.

Performance Efficiency

This focuses on how the system uses its resources. INTRAK is designed to respond smoothly to user actions and handle expected workloads. Extra attention is given to the speed and stability of file uploads and downloads, especially when using Network Attached Storage (NAS).

Usability

Usability refers to how easy the system is to understand and operate. The interface is designed to be simple and clear so that users with different levels of technical knowledge can navigate the system, reduce mistakes, and perform tasks without difficulty.

Reliability

This characteristic deals with the system's ability to run consistently without failure. INTRAK includes backup and recovery measures so that data and services remain available even when unexpected issues occur.

Security

Security ensures that information is protected from unauthorized access. INTRAK uses features such as user authentication, access control, data encryption, and activity logging to keep records safe and maintain accountability.

Maintainability

Maintainability describes how easily the system can be updated or improved. INTRAK follows a modular structure and uses organized documentation and coding practices, making it easier for developers to add new features or fix issues in the future.

Portability

Portability refers to the system's ability to work on different platforms. Since INTRAK is web-based and built using standard technologies, it can be adapted to new environments or systems when needed.

These quality characteristics serve as a foundation for measuring the system's performance and ensuring that INTRAK meets the expectations set for its development and use that follows the mentioned engineering standard

1.7.2 ISO/IEC 27001: Information Security Management

ISO/IEC 27001 is an international standard for managing information security within organizations. It guides institutions in identifying risks, implementing controls, and maintaining secure information management processes. Applicable to any organization, the standard ensures recognized practices for preventing unauthorized access, data loss, or misuse. By combining policies, procedures, and technology, ISO/IEC 27001 helps organizations manage cybersecurity threats, enhance resilience, and maintain secure and stable operations.

INTRAK adopts several core ideas from ISO/IEC 27001 to protect student information and academic records. These principles guide how the system is designed and maintained.

Confidentiality

Access to sensitive information is restricted to authorized users. INTRAK uses defined user roles, login requirements, and controlled permissions to ensure that only individuals with the proper authority can view or manage specific data.

Integrity

The system includes mechanisms that help keep data accurate and unaltered. Input checks, database safeguards, and activity logs are used to prevent unauthorized changes and to track actions taken within the system.

Availability

INTRAK is built to provide users with reliable access to information. Its system setup, backup routines, and server configuration help reduce downtime and ensure that data remains reachable when needed.

1.7.3 Database Management Standards

ACID Properties in PostgreSQL

PostgreSQL follows the ACID principles to keep the data in INTRAK accurate and dependable. These properties are important because the system records attendance, updates documents, and receives evaluation results, all of which must remain correct even when many users are active.

Atomicity

In INTRAK, atomicity ensures that operations such as attendance entries, document status updates, and evaluation submissions are completed as a single unit. If any part of the process fails—such as a network interruption or invalid input—the entire transaction is cancelled and no partial data is saved. This prevents incomplete records that could affect academic monitoring.

Consistency

Consistency allows INTRAK to maintain accurate and valid information throughout the system. All database rules, such as required fields, allowed values, and defined relationships between student records and OJT data, must be met before any transaction is accepted. If an operation would lead to incorrect or conflicting information, it is blocked, helping preserve reliable academic and administrative data.

Isolation

Isolation is important in INTRAK because multiple users—such as coordinators, advisers, and students—often access the system at the same time. Each user's transaction is processed separately, so changes made by one user are not visible to others until the transaction is completed. This prevents issues such as overwritten entries or inconsistent document statuses during simultaneous updates.

Durability

Durability ensures that once INTRAK saves a transaction, such as a finalized attendance log or approved document, the data remains stored even if the system encounters an unexpected shutdown or failure. PostgreSQL writes committed transactions to stable storage, allowing the system to recover without losing confirmed information.

1.7.4 Web Development Standards

W3C Web Content Accessibility Guidelines (WCAG) 2.1

WCAG 2.1 provides guidelines for making web content more accessible to people with disabilities, including visual, hearing, physical, or cognitive limitations. The goal is to ensure websites are accessible on various devices—computers, laptops, and mobile phones—and that users can read, understand, and navigate content. While WCAG 2.1 cannot cover every individual need, following it improves overall usability.

The standard is organized into testable success criteria applicable to any web technology. WCAG 2.1 expands on WCAG 2.0 with updates for modern devices and user requirements. Content that follows WCAG 2.1 also meets WCAG 2.0, and using the newer version is recommended for long-term accessibility and relevance.

INTRAK incorporates key WCAG 2.1 principles to support accessible and user-friendly interaction. The interface uses semantic HTML elements and ARIA labels to help screen readers interpret content accurately. Keyboard navigation is supported through focusable elements and visible focus indicators, assisting users who cannot rely on a mouse. Color choices are selected with contrast considerations, including dark mode support, allowing users with low vision to distinguish elements more easily. These measures help ensure that students, instructors, coordinators and supervisors can access and use the system without difficulty.

1.7.5 Data Privacy Standards

Republic Act No. 10173 - Data Privacy Act of 2012 (Philippines)

The system adheres to the provisions of Republic Act No. 10173, also known as the Data Privacy Act of 2012. Its design applies the core principles required by Philippine data privacy regulations to ensure the safe and lawful handling of personal information within the INTRAK platform. The system includes consent mechanisms through the required Consent Form document submission, limits data use only to approved and stated purposes related to OJT management, and collects only the information necessary for system operations through data minimization practices.

In addition, the system upholds the rights of data subjects by allowing users to view, verify, and directly update their personal data through their profile settings. Users can access their information, review their records, and make corrections to their personal details without requiring administrative intervention. All data processing activities follow the guidelines issued by the National Privacy Commission for educational institutions. These measures help maintain confidentiality, integrity, and responsible management of personal information throughout the system lifecycle.

1.7.6 Network Storage Standards

INTRAK's NAS integration uses standard network file-sharing protocols, specifically SMB and CIFS, to ensure compatibility with existing and future storage systems. The NAS is mounted at the operating system level, allowing the application to access files like a regular filesystem.

The connection between the cloud-hosted application (Hostinger VPS) and the on-campus Raspberry Pi NAS is secured using a Tailscale VPN, which encrypts data and maintains institutional control. SMB and CIFS handle file sharing, access control, and data integrity, providing cross-platform compatibility and supporting system scalability.

1.8 Engineering Design Process

The development of INTRAK follows a structured engineering design process adapted for software systems. It begins with identifying user requirements and defining both functional and non-functional needs, followed by system design, prototyping, and implementation with accordance to engineering practices. Testing ensures functionality, performance, and compliance with data privacy regulations. After deployment, the system is maintained and continuously improved to ensure long-term reliability, efficiency, and adaptability.

Requirements Analysis

The development of the INTRAK system begins with requirements analysis, where the needs of all users, including students, faculty, and administrative staff, are identified. This stage defines both functional requirements, such as document submission, retrieval, and approval workflows, and non-functional requirements, including system security, performance, and interoperability with NAS storage and network protocols.

.System Design

The system design phase establishes the overall architecture of the system, detailing modules, databases, and network components. During this stage, appropriate engineering standards and protocols, such as SMB/CIFS for NAS integration and secure authentication mechanisms, are selected. Considerations for scalability and future enhancements are also incorporated to ensure the system remains adaptable and sustainable.

Prototyping or Modeling

In the prototyping or modeling phase, low-fidelity mockups or working prototypes are created to validate system features. Network interactions and storage operations are simulated, and feedback from stakeholders is gathered to refine the design and address any potential issues before full implementation.

Implementation or Coding

The implementation or coding phase involves developing the application using the chosen programming languages and frameworks. This stage includes integrating NAS storage, user authentication, and workflow automation while adhering to engineering best practices such as modularity, proper documentation, and version control to ensure a maintainable and reliable system.

Testing and Validation

During testing and validation, the system is evaluated to ensure that all functional and non-functional requirements are met. Functional testing focuses on document submission, retrieval, and approval workflows, while performance testing examines file transfer speeds and system response times. Compliance with data privacy regulations, particularly RA 10173 (Data Privacy Act), is verified to protect sensitive information and prevent unauthorized access.

Deployment

In the deployment phase, the system is deployed on a cloud-hosted Virtual Private Server (VPS) provided by Hostinger, with the application and database hosted in the cloud for global accessibility and scalability. The Raspberry Pi NAS storage located within the campus network is connected to the cloud application server via a Tailscale VPN tunnel, enabling secure document storage while maintaining institutional control over sensitive data. Training is provided to end-users and administrators to ensure effective system usage and smooth operational transition from traditional paper-based processes to the digital platform.

Operation and Maintenance

The operation and maintenance phase involves monitoring system performance, applying updates and patches, and addressing any issues that arise during regular use. Continuous feedback is collected to identify areas for improvement and ensure the system remains functional and efficient.

Evaluation and Continuous Improvement

Finally, evaluation and continuous improvement are conducted to review system performance and user satisfaction. Adjustments are made based on evolving institutional requirements, and lessons learned are documented to guide future engineering projects and ensure the system's long-term reliability and adaptability

1.8.2 Research the Problem

The team analyzed commercial OJT management platforms to understand common feature sets and architectural patterns, while recognizing that standard solutions did not adequately address the specific requirements and constraints of the Computer Engineering Department in PSU–Urdaneta City Campus.

Technical research was conducted to compare different storage options, including NAS, cloud, and local servers, as well as database systems suitable for managing academic records. Web development frameworks were also evaluated for their security, performance, and ease of maintenance. Insights from software engineering literature guided the choice of system architecture and the development approach.

Regulatory research examined data privacy requirements under Philippine law, document retention policies for educational institutions, and accessibility standards for public sector web applications. This ensured that the system's design would comply with legal and institutional requirements while remaining secure and user-friendly.

1.8.3 Imagine: Develop Possible Solutions

The researchers identified and conceptualized possible system design solutions that could address the storage, security, and operational requirements of the INTRAK System.

Three distinct system design approaches were formulated to represent different ways of managing internship data and system infrastructure. These include (1) a cloud-based architecture with NAS integration using a Raspberry Pi, (2) a fully cloud-based architecture without NAS integration, and (3) a cloud-based architecture with commercial NAS integration. Each approach was designed to address the same system objectives but through different technical implementations.

Tool that can be used for System Development:

CATEGORY	Tools
Frontend Framework	React 19 + Typescript Vue.js 3 / Next.js +Type Script / Angular 17
Styling Framework	Tailwind CSS Bootstrap 5 / Material UI
State Management	React Context API +Hooks Redux Toolkit / Zustand Pinia / (Redux/Conext)
Routing	React Router v7 Angular Router next.js router / tanStack router Vue Router
HTTP Client	Axios Angular HttpClient

	Fetch API / SWR
Form Handling	React Hook Form + Zod Angular Reactive Forms Formik + Yup / React Final Form
Build Tool	Vite Maven/Gradle + Angular CLI Create React APP / Webpack
Backend Runtime	Node.js 22 LTS Java Runtime Environment (JRE) / JVM Python 3.x Runtime
Backend Framework	express.js Java (Spring Boot) Python (Django /Flask) nestJS / Fastify
Backend Language	Typescript Java Javascript / Python
ORM/ Database Tool	Prisma Spring Data JPA / HIbernate Sequelize / Type ORM Django ORM / SQLAlchemy
Database	Postgres SQL MySQL /MongoDB
Authentication	jsonwebtoken passport.js / Auth0
File Upload	Multer Spring Boot MultipartFile / Cloud Storage NAS Formidable / Busboy Python shutil / SMB Protocol
Email Service	Nodemailer + SendGrid JavaMailSender AWS SES / Mailgun Django Mail / Flask-Mail
PDF Generation	PDF kit iText / Apache PDFBox Puppeteer / jsPDF

	ReportLab / WeasyPrint
Excel Generation	Excel.js Apache POI sheetJS / node-xlsx Pandas / OpenPyXL
Real-Time Communication	Socket.io WebSockets / Server-sent events
Validation Library	Zod HIbernate Validator / Jakarta Bean Validation Yup / Joi
Testing Framework	Jest + PyTest JUnit & Mockito + Jasmine & Karma Vitest / Mocha + Cho
Code Quality	ESlint + Prettier SonarQube / Checkstyle StandardJS/ Biome

The INTRAK system can be developed using a variety of modern tools and frameworks to ensure a robust, scalable, and maintainable architecture. For the frontend, options such as React 19 with TypeScript, Vue.js 3, or Angular 17 provide a dynamic and responsive user interface. Styling can be implemented using Tailwind CSS, Bootstrap 5, or Material UI, while state management can be handled using React Context API with Hooks, Redux Toolkit, or Zustand. Routing and HTTP requests are supported through tools like React Router v7, Next.js Router, Axios, or the Fetch API, and form handling can leverage React Hook Form with Zod, Formik with Yup, or React Final Form. The frontend build process can utilize Vite, Create React App, or Webpack.

For the backend, runtime environments such as Node.js 22 LTS, Python (Django/Flask), or Java (Spring Boot) can be employed, with frameworks like Express.js, NestJS, or Fastify to structure the application. Backend development can be done using TypeScript, JavaScript, or Python, while database operations can utilize PostgreSQL, MySQL, or MongoDB alongside ORMs such as Prisma, Sequelize, or TypeORM. Additional tools include jsonwebtoken, passport.js, or Auth0 for authentication; Multer, Formidable, or Busboy for file uploads; Nodemailer with SendGrid or AWS SES for email; and PDFKit, Puppeteer, or jsPDF for PDF generation. Real-time communication is facilitated by Socket.io, WebSockets, or Server-Sent Events, while validation can use Zod, Yup, or Joi. Testing and code quality are supported through Jest, Vitest, Mocha + Chai, and ESLint with Prettier.

1.8.4 Plan: Select a Promising Solution

The researchers developed and evaluated three alternative system designs to determine the most suitable solution for the INTRAK system. Each design was carefully assessed based on important factors such as cost, data security, system performance, scalability, and overall suitability to the institution's needs.

The researchers concluded that Design 1 (cloud-based architecture with NAS integration using a Raspberry Pi) is the most suitable solution for the INTRAK system. It effectively addresses the system requirements, supports future scalability, minimizes operational costs, and ensures secure handling of internship records. These advantages make it the most practical, reliable, and efficient design for the successful implementation of the INTRAK system.

1.8.5 Create: Build a Prototype

The researchers developed a functional prototype of the INTRAK System to demonstrate the feasibility of the proposed cloud-based architecture with Raspberry Pi NAS storage. The prototype served as an initial working model of the system, allowing verification of the design, evaluation of core functionalities, and validation of the workflow for different user roles.

Prototype Architecture and Components

The prototype followed the design outlined in the previous sections, consisting of the following key components:

Cloud-Based: Hosted on a Hostinger VPS, the web application serves as the interface for all users. The application provides modules for students, instructors, coordinators, administrators, and host institutions.

NAS Storage (Raspberry Pi 4): Sensitive documents uploaded by users are stored locally on a Raspberry Pi 4 configured as a NAS. The connection between the cloud application and NAS is secured through a Tailscale VPN, ensuring safe file transfer.

Database: A Dockerized PostgreSQL database on the VPS stores metadata for all uploaded documents, user information, and activity logs.

Development Process

The prototype was built using an iterative development approach, progressing through several cycles:

Foundation Stage:

1. Database tables for users, documents, and metadata were created.
2. User roles and permissions were configured to differentiate between students, coordinators, administrators, and host institutions.
3. The NAS system was integrated to handle file storage, with test uploads to verify secure storage and retrieval.
4. Basic login and authentication features were implemented to ensure that access control worked as intended.

Feature Implementation Stage:

1. Student Module: Students could register, log in, and submit internship documents through the dashboard.
2. Coordinator Module: Coordinators could review submissions, approve documents, and provide feedback.
3. Attendance Tracking: A simple module was implemented to monitor student internship attendance.
4. Reporting Module: The initial version allowed administrators to generate summary reports of document submissions and student progress.
5. Integration and Testing:
6. Each module was tested individually to ensure functionality.

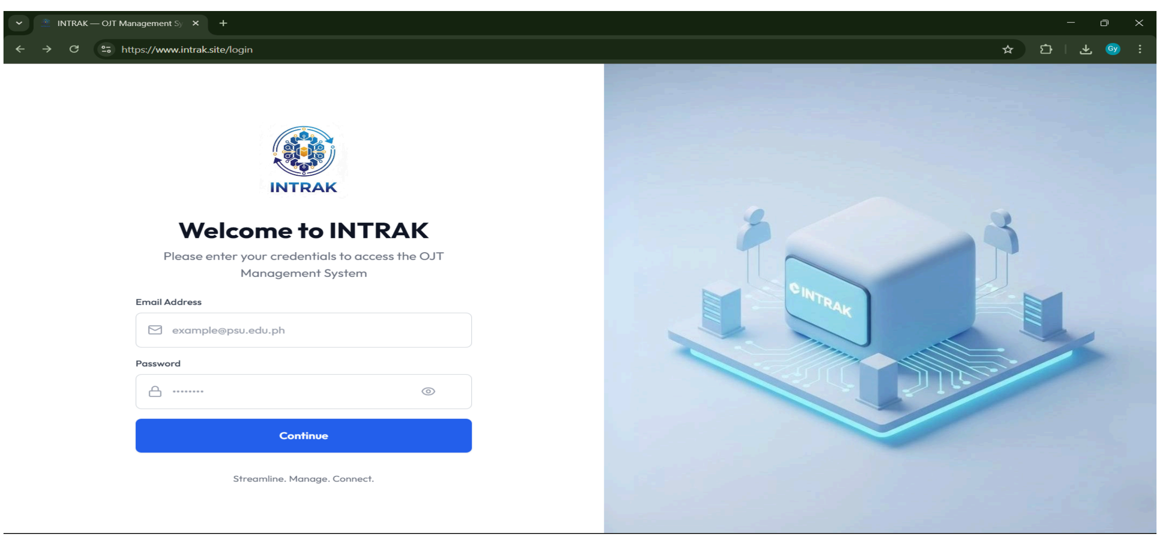
7. End-to-end workflow tests were performed, such as uploading a document as a student, storing it securely in the NAS, and accessing it as a coordinator.
8. Role-based access was validated to confirm that each user could only access permitted features and data.

Prototype Dashboard

The main interface of the prototype is the INTRAK Dashboard, which allows users to interact with the system efficiently.

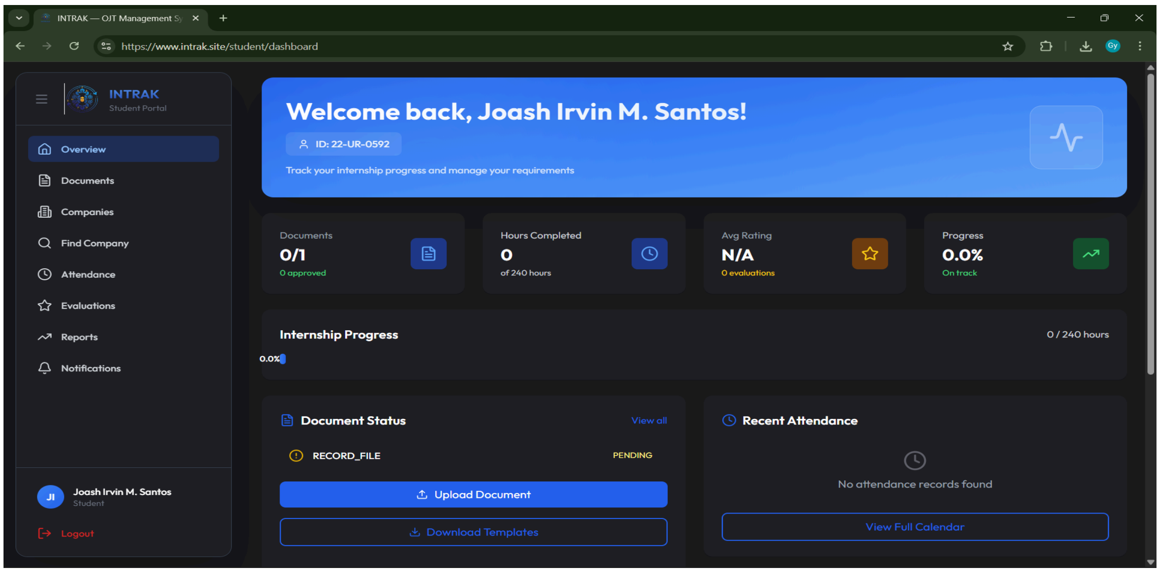
LOG IN INTERFACE:

Access to the system is secured through a login interface where students, instructors, coordinators, and supervisors enter their credentials. Each user is directed to a role-specific dashboard that provides the necessary tools and information relevant to their responsibilities while maintaining secure, controlled access to sensitive internship data.



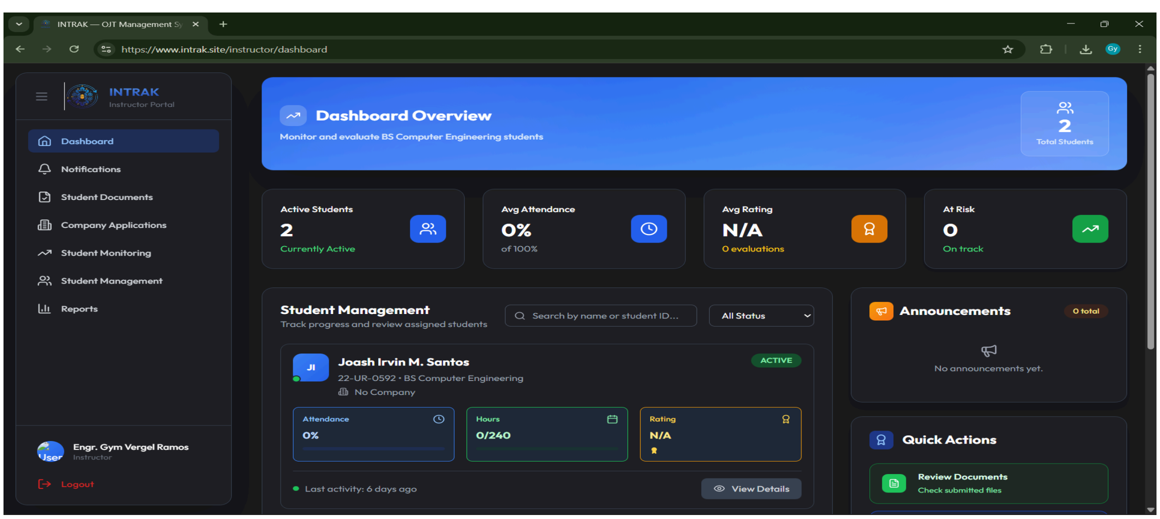
STUDENT DASHBOARD

It allows students to submit digital copies of internship requirements such as Daily Time Records (DTR), performance evaluations, and other related documents. Students can monitor the status of their submissions in real-time and receive feedback from coordinators, which helps them stay informed, organized, and compliant with internship requirements.



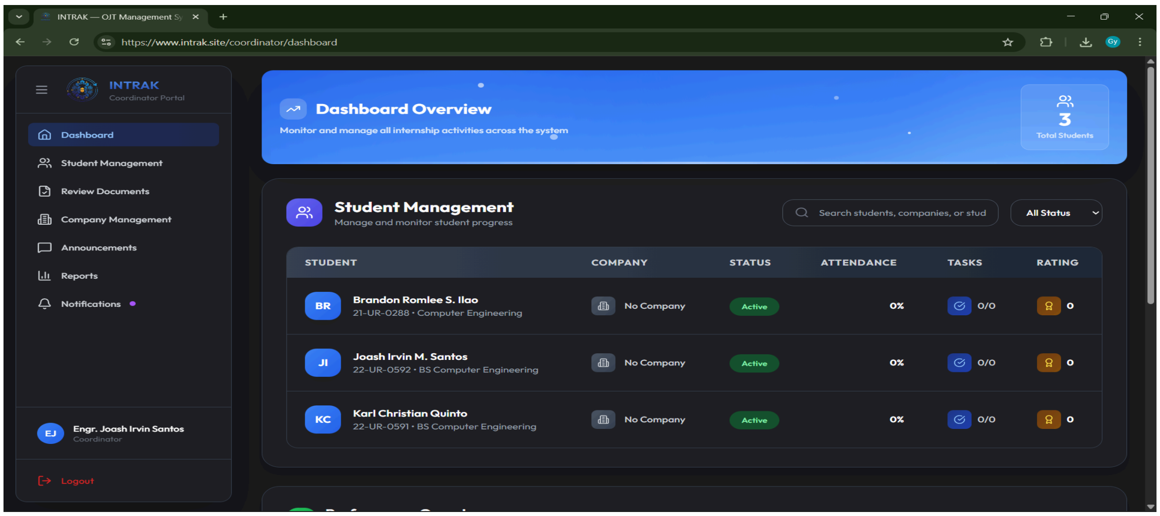
INSTRUCTOR DASHBOARD

Instructors have the capability to review student submissions, validate uploaded documents, and give feedback. Instructors can also monitor student attendance through login timestamps and track internship progress, ensuring that students complete their tasks according to university guidelines.



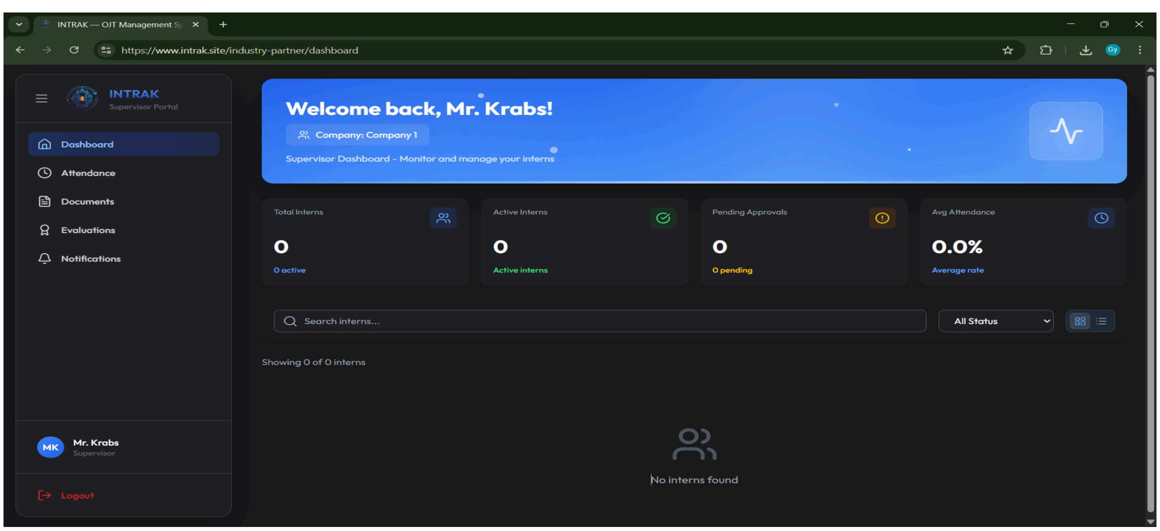
COORDINATOR DASHBOARD:

The coordinator dashboard will serve as the administrative hub for internship monitoring. Coordinators can approve submissions, generate reports for students, instructors, and the university, and evaluate overall student performance. The tools provided help streamline workflow, reduce manual monitoring, and maintain proper documentation for assessment purposes.



SUPERVISOR DASHBOARD

The supervisor dashboard is designed for host institution supervisors to verify attendance, confirm completion of tasks, and submit performance evaluations. This feature ensures clear communication between the university and internship sites, centralizing the oversight process while maintaining accountability.



By the end of the prototype phase, the researchers successfully developed a working model of the INTRAK System. The prototype demonstrated the feasibility of the cloud-based + NAS architecture for secure and cost-efficient document storage.Proper implementation of role-based access controls for different user types.Smooth workflow for document submission, verification, and reporting.The prototype served as a baseline for further enhancement and provided a clear reference for the full-scale development and deployment of the INTRAK System. It also allowed the researchers to identify potential improvements before final implementation, ensuring that the final system would meet operational requirements effectively.

1.8.6 Test and Evaluate Prototype

Testing activities encompassed multiple levels to ensure system quality and reliability. Unit tests validated individual functions and components in isolation, while integration tests verified correct interaction between system modules. User interface testing assessed usability, browser compatibility, and responsive design effectiveness.

User acceptance testing was done with real users, including students, instructors, and coordinators, using a test version of the system that worked like the final system. During these tests, users performed normal tasks to see how the system worked in practice. Their feedback helped identify ways to make the system easier to use and improve workflow. The testing also confirmed that the system met the needs of the users and the institution.

Performance testing will be conducted using students from the Computer Engineering Department, OJT coordinators, and assigned coordinators from partner companies. These tests will conduct a simulation with the help of selected students, instructors, coordinators and supervisors to check if the system can handle multiple users and document uploads at the same time without slowing down. Security testing, including checking for vulnerabilities and attempting simulated attacks, will evaluate how well the system protects data and controls access through authentication and authorization mechanisms.

1.8.7 Improve: Redesign as Needed

The improvement phase encompasses both pre-deployment refinements based on testing feedback and planned post-deployment enhancement cycles. Results show specific improvements including interface refinements for mobile browsers, optimization of database queries for report generation, and strengthening of file validation logic for document uploads.

The INTRAK system is developed using an iterative engineering design process, which allows for continuous improvement even after the system is first deployed. After implementation, the system will be monitored, and feedback from users will be collected. Regular reviews will help identify opportunities to enhance the system, add new features, and optimize performance in future updates. This iterative approach ensured that INTRAK development continued in multiple sprint cycles, allowing continuous improvement even after deployment.